Req ID	Requirement	Traced To (Use Case)	Traced To (Component)	Verification Method	Status
FR-001	Ingest consumption logs and compute CO2e emissions	Log Monthly Consumption; Compute CO2e Emissions (<<include>>)	Consumption & Target Manager; CO2e Conversion Service	Unit test: conversion formula vs. EPA factors	Verified
FR-002	Set a monthly/annual carbon-reduction target	Set Reduction Target	Consumption & Target Manager	Functional test: target persistence across sessions	Verified
FR-003	Calculate and display progress toward active target	Track Progress Toward Target	Consumption & Target Manager	Functional test: progress % recalculation after log	Verified
FR-004	Notify Resident when a progress milestone is reached	Receive Milestone Alert (<<extend>>)	Notification & Coach Review	Functional test: single alert per milestone threshold	Verified
FR-005	Coach views household data and submits recommendations	Review Household Data; Provide Recommendations	Notification & Coach Review	Functional test: recommendation scoped to correct household	Verified
NFR-001	Milestone chart renders in under 200ms under peak load	View Historical Comparison Chart	User Interface Component; Database Component	Performance benchmark under simulated peak load	Verified
NFR-002	Household data encrypted and access-scoped	(cross-cutting — all use cases touching household data)	Database Component	Security penetration test; AES-256 verification	Verified